

Topic: Memory Segmentation in 8086

Q. Explain the concept of Memory Segmentation in the 8086 microprocessor. How is a 20-bit physical address calculated using a 16-bit segment register and a 16-bit offset?

****Definition to Remember****

****Memo**
internal
to gener

Memory Segmentation in 8086

The CPU manages these memory blocks using four specialized 16-bit Segment Registers:

1. Code

2. Logical vs Physical Address

* ****Logical Address:**** Format used by programmers `(Segment : Offset)`. e.g., `1000H:0002H`.

* ****Ph**

3. Calculating the 20-bit Physical Address

The Bus Interface Unit (BIU) performs this calculation:

Formul

Calculation Example:

* `CS`

1. Shift Segment by 10H: `2000H x 10H` = `20000H` (20-bit Base)

2. Add

4. Advantages of Segmentation

* ****Code Relocation:**** Programs can be moved anywhere in memory by changing only the Segment Register; internal offset addresses remain unchanged.

* ****M**
overwri

****Must-Write Points (for 10 marks)****

1. 8086

Quick Recall

`Segme
10H) + c